UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

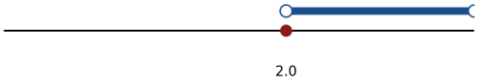
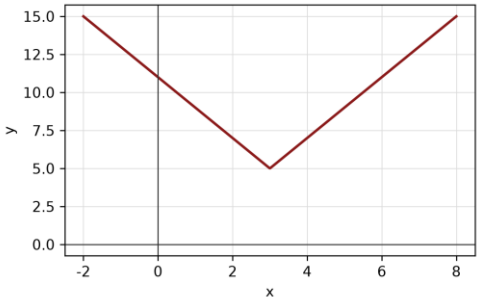
INSTRUCTIONS TO MARKERS

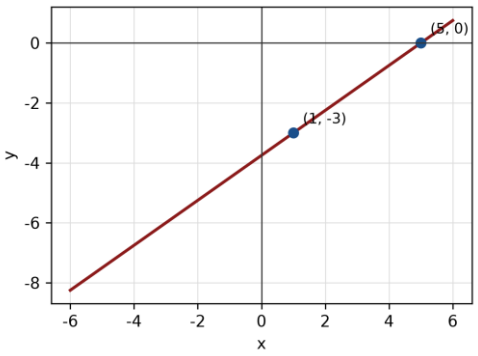
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

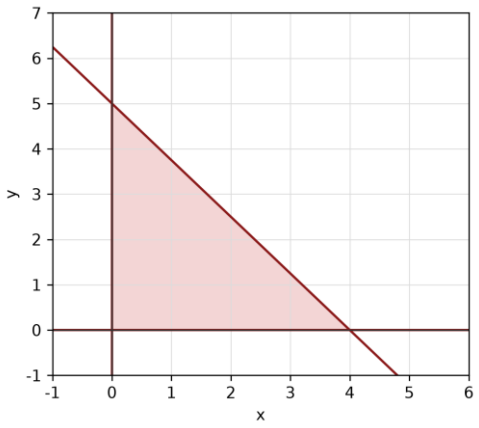
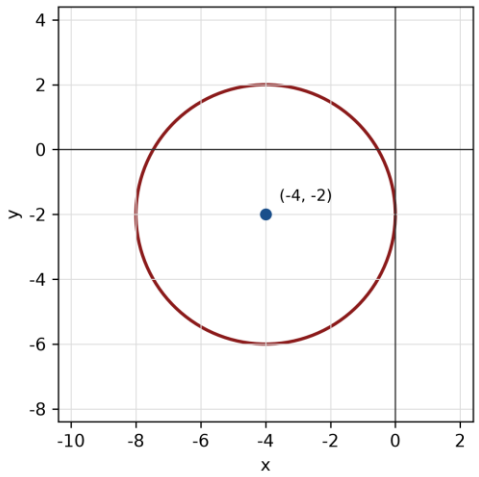
FOR MARKERS' USE ONLY

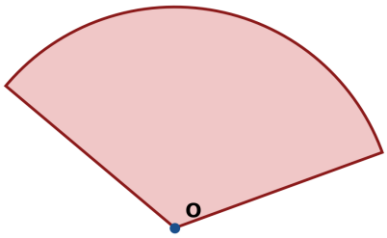
This marking scheme consists of 5 printed pages

«:> Hak Cipta Universiti Teknologi MARA

NO	SOLUTION		MARK
1 a)	$2-4x<-6$ $-4x<-8$ $(2,\infty)$  	M1 A1	4 m
1 b)	$\lvert 5x+2\rvert-3<0$ $-3<5x+2<3$ $\left(-1,\frac{1}{5}\right)$ 	M1 A1 A1	5 m
1 c)	$y=2\lvert x-3\rvert+5$ vertex $(3,5)$ no real intercepts  	B1 A1 A1	5 m
2 a)	$(g\circ f)(x)=g(f(x))$ $g(2x-6)=-\frac{5}{2x-11}$ $x\neq\frac{11}{2}$	M1 A1 B1	5 m

NO	SOLUTION		MARK
2 b)	$1x + 3 \neq 0$	M1	4 m
	$x \neq -3$	A1	
	$3x + 3 \geq 0$	M1	
	$x \geq -1$	A1	
2 c)	$m = \frac{3}{4}$	M1	4 m
	$y + 3 = \frac{3x}{4} - \frac{3}{4}$	M1	
	$y = \frac{3x}{4} - \frac{15}{4}$	A1	
			
3 a)	$(2 - 4i) - (5 + i)$	M1 A1	4 m
	$-3 - 5i$	A1	
3 b)	$i^{33} = i^{4(8)+1}$	M1	3 m
	$= (i^4)^8 i^1 = 1^8 i^1$	A1	
	$= i$	A1	
3 c)	$\log_2 [x(x - 14)] = 5$	M1 M1	5 m
	$x(x - 14) = 32$	M1	
	$x^2 - 14x - 32 = 0$	A1	
	$x = 16$	A1	

NO	SOLUTION	MARK
4 a)	$x+8=x^2-4$ M1 $x^2-x-12=0$ A1 $(x-4)(x+3)=0$ $x=4$ or $x=-3$ $(4,12), \quad (-3,5)$ A1	5 m
4 b)	$\frac{x}{4}+\frac{y}{5}=1$ M1 A1 $x\geq 0, \quad y\geq 0, \quad 5x+4y\leq 20$ A1 	5 m
4 c)	$(x+4)^2+(y+2)^2=16$ M1 $x^2+8x+y^2+4y+20=16$ A1 $x^2+8x+y^2+4y+4=0$ B1 	5 m

NO	SOLUTION		MARK
5 a)	$\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$	M1	5 m
	$\sin^2(x) = 1-\cos^2(x)$	M1	
	$\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$		
	$\frac{\sin^2(x)}{1-\cos(x)} = \cos(x)+1$	A1	
5 b)	$s = r\theta = 6\pi$	M1 A1	4 m
	$P = 2r+s = 18+6\pi = 18+6\pi$	A1	
			
5 c)	$A_{\triangle} = \frac{1}{2}(9)(8) = 36$	M1	5 m
	$A_{\text{sector}} = \frac{1}{2}(3)^2\left(\frac{90\pi}{180}\right) = \frac{9\pi}{4}$	M1	
	$A_{\text{shaded}} = 36-\frac{9\pi}{4} = 36-\frac{9\pi}{4}$	A1	

«:> Hak Cipta Universiti Teknologi MARA